

51.504 Machine Learning (2021)
Final

2 hours

Question 1 [20 points]

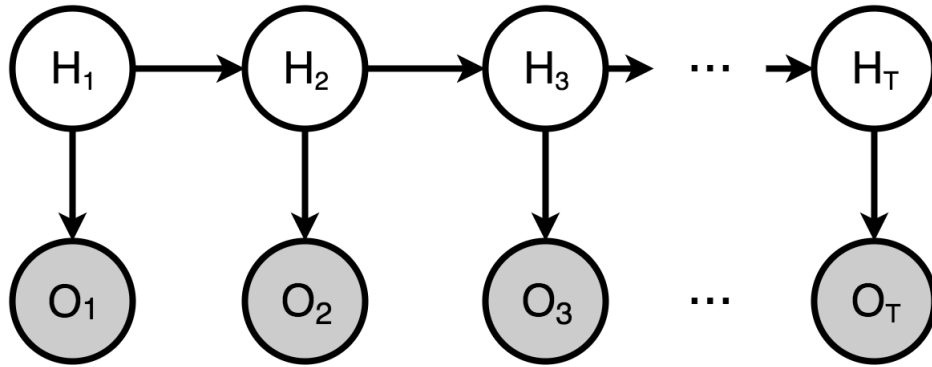


Figure 1: Hidden Markov Model G

Consider the Hidden Markov model above.

- (a) (i) [4 points] Name the algorithm used to learn the most likely state of H_k , $1 \leq k \leq T$, given the observed states? Are the parameters known?
- (ii) [4 points] Name the algorithm used to learn the most likely hidden sequence $H_1, \dots, H_T$ given the observed states? Are the parameters known?
- (iii) [2 points] In parameter learning, the emission parameters are updated in the E-step or the M-step?
- (iv) [2 points] Name the algorithm used to compute the conditional probabilities in the E-step.
- (b) Now let the states of H_i be $\{1, 2, 3\}$, and the states of O_i be $\{\text{"the"}, \text{"cat"}, \text{"jumps"}, \text{"up"}\}$. Assume that we initialize the transition parameters such that $p_{31} = P(H_t = 1 \mid H_{t-1} = 3)$ and $p_{32} = P(H_t = 2 \mid H_{t-1} = 3)$ are 0, and all the others are positive.

After several iterations of the EM algorithm, we find that the emission parameters have been updated to be:

$$P(O_t = \text{"the"} \mid H_t = 1) = 0.63$$

$$P(O_t = \text{"cat"} \mid H_t = 1) = 0.37$$

$$P(O_t = \text{"up"} \mid H_t = 2) = 0.89$$

$$P(O_t = \text{"the"} \mid H_t = 2) = 0.11$$

$$P(O_t = \text{"jumps"} \mid H_t = 3) = 0.14$$

$$P(O_t = \text{"cat"} \mid H_t = 3) = 0.86$$

- (a) [4 points] Given that $P(H_1 = 2) = 0.3$ and $P(H_2 = 2 \mid H_1 = 2) = 0.4$, calculate the probability $P(O_2 = \text{"up"} \mid H_2 = 2, H_1 = 2)$.
- (b) [4 points] At this point, which of the following output sequences is definitely impossible? Explain why.
- (1) "up", "the", "cat", "jumps"
 - (2) "the", "cat", "jumps", "up"
 - (3) "the", "cat", "the", "cat"
 - (4) "cat", "up", "jumps", "cat"

Question 2 [18 points]

Suppose we use the EM algorithm to perform clustering ($K = 2$) on a sample dataset with 4 points $\{x^{(1)} = 1, x^{(2)} = -1, x^{(3)} = 2, x^{(4)} = -3\}$, all lying on the real line. After the E-step we have

	$x^{(1)}$	$x^{(2)}$	$x^{(3)}$	$x^{(4)}$
$\gamma(z_1^{(i)}) = P(Z = 1 x^{(i)})$	0.3	0.2	0.6	0.9
$\gamma(z_2^{(i)}) = P(Z = 2 x^{(i)})$	0.7	0.8	0.4	0.1

(a) Use the formulas for the M-step

$$\pi_k = \frac{1}{N} \sum_{i=1}^N \gamma(z_k^{(i)}),$$

$$\mu_k = \frac{\sum_{i=1}^N x^{(i)} \gamma(z_k^{(i)})}{\sum_{i=1}^N \gamma(z_k^{(i)})},$$

$$\sigma_k^2 = \frac{\sum_{i=1}^N (x^{(i)} - \mu_k)^2 \gamma(z_k^{(i)})}{\sum_{i=1}^N \gamma(z_k^{(i)})},$$

to compute

- (i) [4 points] π_1, π_2 ,
- (ii) [4 points] μ_1, μ_2 ,
- (iii) [4 points] and σ_1^2, σ_2^2

for the next iteration. For (iii), you do not need to evaluate the final expressions if you do not have a calculator.

(b) Answer the following statements with True or False:

- (i) [2 points] The EM algorithm uses hard assignments to cluster the data-points.
- (ii) [2 points] The steps for updating the parameters in the EM algorithm are derived from the closed-form solution of maximizing the log-likelihood of the Gaussian mixture model.
- (iii) [2 points] The proportion of clustering in EM is represented by $\gamma(z_k^{(i)})$.

Question 2 [18 points]

Question 3 [18 points]

Refer to the directed acyclic graph H in Figure 2. Let the letters $\{A, \dots, G\}$ refer as well to the joint random variable represented by this Bayesian network.

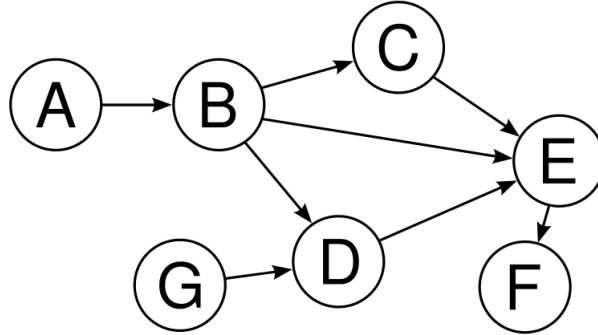


Figure 2: Directed Graph H

- (a) [3 points] If this was a directed Gaussian graphical model, which node would present a $N(0, 1)$ distribution? Explain.
- (b) [3 points] Write down the local Markov property with respect to node D .
- (c) [3 points] Is node D d-separated from C ?
- (d) [3 points] Is node D d-separated from C given $\{B, F\}$?
- (e) [3 points] Write down a conditional independence relationship read off the graph using the global Markov property.
- (f) [3 points] Write the factorization for $P(A, B, C, D, E, F, G)$.

Question 3 [18 points]

Question 4 [18 points]

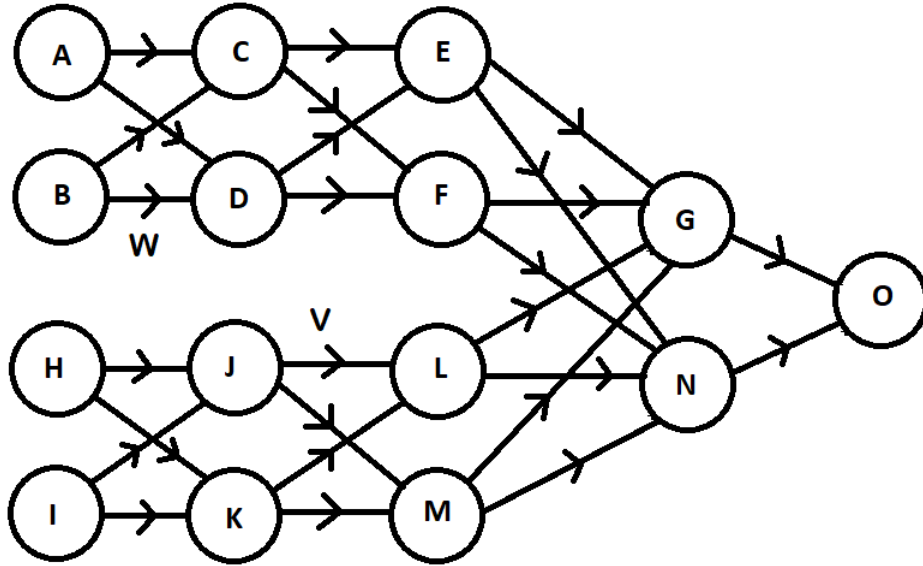


Figure 3: Neural network

- (a) (i) [2 points] Does the neural network in Figure 1 represent a directed acyclic graph (DAG)?
- (ii) [4 points] Write down every node that has an input that contains the output of node J that passes through V in the feed forward process.
- (iii) [4 points] The error signal δ_O at the output node O has just been computed. To compute the gradient of the loss function with respect to the weight W , what is the minimal set of error signals that has to be computed? Write down the error signals in the correct order starting with δ_O ; eg. if you believe that next the error signal should be computed at N , followed by M , ... , write $\delta_O, \delta_N, \delta_M, \dots$.
- (b) (i) [4 points] What is the output when we convolve the left image with the 2×2 filter on the right, with stride 1, padding 1 and applying ReLU?

0	7	0
1	3	2
2	1	5

*

0	-1
1	0

- (ii) [4 points] What is the result if we apply max pooling with a 2×2 filter and stride 2 to the above output?

Question 4 [18 points]

Question 5 [12 points]

Determine if the following statements regarding reinforcement learning are true or false:

- (a) [2 points] The Upper Confidence Bound method is stochastic. _____
- (b) [2 points] The ϵ -greedy method does not choose intelligently between the actions that are not exploited. _____
- (c) [2 points] Maximizing the immediate expected reward at each step will maximize the long term expected reward. _____
- (d) [2 points] Hoeffding's inequality allows us to determine the appropriate ratio between exploitation and exploration so that to guarantee increase in the expected reward over time. _____
- (e) [2 points] Policy is a deterministic plan of action for the agent given states. _____
- (f) [2 points] The value of a state is the total amount of reward that an agent can expect to get from that state in the long run. _____

Question 6 [14 points]

Determine if the following statements are true or false:

- (a) [2 points] The likelihood function can be expressed as a product of functions of individual samples because the samples are independent and identically distributed. _____
- (b) [2 points] Gradient descent is required to solve for the best linear estimate in linear regression. _____
- (c) [2 points] The perceptron algorithm is a mistake driven algorithm. _____
- (d) [2 points] The K-means algorithm is only possible to implement if we initialize it with a set of centroids first. _____
- (e) [2 points] The solution for linear regression can be expressed in terms of a kernel. _____
- (f) [2 points] For functions that have no analytic expression, stochastic gradient descent can be used to find optimal points. _____
- (g) [2 points] Genetic algorithms make use of random selection to get out of local optimal points in optimization. _____